MACHINE LEARNING
STUDY MATERIAL

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Department	CSE AI/ML
Programme	B.Tech
Study focus	KTU-aligned introductory Machine Learning concepts

A concise student-friendly revision guide covering core ideas, algorithms, evaluation, and practical examples.

Prepared for learning and revision. Always compare topics with the latest syllabus and course notes issued by your college.

1. Introduction to Machine Learning

Foundation concepts commonly introduced in KTU AI/Data Science and Machine Learning coursework.

What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence that enables a computer system to learn patterns from data and use those patterns to make predictions or decisions. Instead of writing every rule manually, developers provide examples, choose a suitable algorithm, and evaluate the learned model.

Why is ML useful?

• **Prediction:** estimate a future or unknown value, such as house price or energy demand.

• **Classification:** assign a label, such as spam/not spam or defective/acceptable.

• **Grouping:** discover natural groups in data, such as customer segments.

• **Automation and discovery:** detect patterns in images, text, sensor readings, and transactions.

Important terms

Term	Meaning
Dataset	A collection of examples used to analyse or train a model.
Observation / sample	One row or instance in a dataset.
Feature	An input variable used by the model, such as area or age.
Label / target	The outcome the model should predict in supervised learning.
Model	A learned mathematical or computational mapping from inputs to outputs.
Training	The process of adjusting model parameters using data.

Basic workflow: define the problem → collect data → clean and explore data → split data → train a model → evaluate it → improve and deploy it.

Example: A model can learn the relationship between house area, location, and sale price from past property records. Its predictions are useful only if the training data is relevant and the model is tested on unseen examples.

2. Types of Machine Learning

1. Supervised learning

The training dataset contains input features and known target labels. The model learns a relationship that can be used to predict targets for new inputs.

• **Regression:** predicts a continuous numeric value. Examples: house price, temperature, project cost.

• **Classification:** predicts a category. Examples: spam detection, disease screening, image class.

2. Unsupervised learning

The data has no supplied target label. The algorithm searches for structure or patterns within the inputs.

• **Clustering:** groups similar observations, for example customers with similar buying behaviour.

• **Dimensionality reduction:** represents data using fewer variables while preserving important information; PCA is a common method.

3. Reinforcement learning

An agent interacts with an environment, takes actions, and receives rewards or penalties. It learns a policy intended to maximise long-term reward. Examples include game-playing agents and robot navigation.

Comparison at a glance

Type	Training signal	Typical task	Example
Supervised	Known labels	Predict output	Predict exam score
Unsupervised	No target labels	Find structure	Group documents
Reinforcement	Reward from actions	Learn a policy	Game-playing agent

Exam tip: Be able to define all three learning types, distinguish regression from classification, and give one realistic example for each.

A model should be chosen based on the task, available data, evaluation requirements, and the cost of incorrect predictions—not simply because an algorithm is popular.

3. Data Preparation and Model Workflow

Data quality strongly affects model quality. Before training, inspect the dataset and make sure the features and target match the problem being solved.

Step 1 — Collect and understand data

• Identify the prediction goal, data source, feature definitions, units, and time period.

• Inspect the number of rows and columns, data types, missing values, duplicates, and unusual values.

• Use descriptive statistics and visualisations to understand distributions and relationships.

Step 2 — Clean and transform

• Handle missing values using a justified method, such as imputation or removing unsuitable records.

• Check outliers; correct data errors rather than automatically deleting every extreme value.

• Encode categorical variables where needed and scale numeric features for algorithms sensitive to scale.

• Prevent **data leakage**: information unavailable at prediction time must not influence training.

Step 3 — Split the dataset

• **Training set:** used to fit model parameters.

• **Validation set:** used to compare models and tune settings.

• **Test set:** held back for a final, relatively unbiased performance estimate.

Step 4 — Train, evaluate, improve

Train a simple baseline first. Compare it with alternatives using metrics suitable for the task. Improve features and model settings using training/validation data, then evaluate once on the test set. After deployment, monitor changes in data and performance.

Common sources of trouble

• Too little data or data that does not represent real-world cases.

• Incorrect labels, inconsistent formats, or unbalanced classes.

• Training and test data that are not separated correctly.

• Using an unsuitable metric or optimising only for accuracy.

Remember: A careful data pipeline is often as important as the choice of algorithm.

4. Regression: Predicting Numeric Values

Regression models estimate a continuous target. Linear regression is a useful starting point because it is simple to interpret and provides a baseline for many prediction problems.

Simple linear regression

The model represents the relationship between one input x and target y as: ■ = b■ + b■x. Here, ■ is the predicted value, b■ is the intercept, and b■ is the slope. For example, x could be house area and ■ the predicted house price.

Multiple linear regression

With several features, the model can be written as ■ = b■ + b■x■ + b■x■ + … + b■x■. Features might include area, number of rooms, age, and location indicators. The coefficients describe the model's fitted relationship, but do not automatically prove causation.

Least-squares objective

Linear regression commonly chooses coefficients that minimise the sum of squared residuals: **SSE =** Σ**(y■ –** ■■**)²**. A residual is the difference between an observed value and its prediction. Squaring makes large errors more influential and prevents positive and negative errors from cancelling.

Gradient descent — core idea

• Start with initial parameter values.

• Calculate the loss on training data.

• Compute the gradient, which indicates the direction of steepest increase in loss.

• Update parameters in the opposite direction: θ **:=** $\theta - \alpha\nabla$**J(**θ**)**, where α is the learning rate.

• Repeat until the loss stops improving or a stopping condition is reached.

Evaluation measures for regression

Metric	Interpretation
MAE	Mean absolute error; average magnitude of errors in target units.
MSE	Mean squared error; penalises large errors more strongly.
RMSE	Square root of MSE; expressed in the target's units.
R^2	Measures fit relative to a mean-target baseline; can be negative on test data.

Worked idea: If actual cost is ■12 lakh and the model predicts ■11 lakh, the prediction error is ■1 lakh. Evaluate this error across many unseen examples before relying on the model.

5. Classification and Logistic Regression

Classification predicts a discrete class rather than a continuous number. Examples include whether an email is spam, whether a component is defective, or which category an image belongs to.

Logistic regression

Despite its name, logistic regression is commonly used for classification. It first forms a linear score z = b■ + b■x■ + … + b■x■ and converts it into a probability using the sigmoid function: **σ(z) = 1 / (1 + e■■)**. The output lies between 0 and 1.

A threshold, often 0.5, can convert the probability into a class. The threshold should be selected according to the application: missing a dangerous defect may be more costly than investigating a false alarm.

Confusion matrix

	Predicted positive	**Predicted negative**
Actual positive	True Positive (TP)	False Negative (FN)
Actual negative	False Positive (FP)	True Negative (TN)

Important classification metrics

• **Accuracy:** (TP + TN) / (TP + TN + FP + FN). Useful when classes are reasonably balanced and error costs are similar.

• **Precision:** TP / (TP + FP). Of predicted positives, how many were truly positive?

• **Recall (sensitivity):** TP / (TP + FN). Of actual positives, how many were found?

• **F1-score:** harmonic mean of precision and recall.

Overfitting and regularisation

Overfitting occurs when a model fits training details too closely and performs poorly on new data. Regularisation discourages overly complex parameter values. L1 regularisation can drive some coefficients to zero; L2 regularisation shrinks coefficients. Validation data helps compare settings.

Exam tip: Know the sigmoid function, the confusion-matrix terms, and when precision or recall matters more than accuracy.

6. Unsupervised Learning, Neural Networks and PCA

Clustering with K-means

K-means partitions observations into K clusters. It repeatedly assigns each point to the nearest cluster centroid and recalculates each centroid as the mean of its assigned points. The process continues until assignments or centroids stabilise.

• Choose K, initialise centroids, assign points, update centroids, and repeat.

• It is efficient for many numeric datasets but sensitive to scaling, initialisation, outliers, and the choice of K.

• Use domain understanding and suitable validation measures to assess whether the clusters are meaningful.

Principal Component Analysis (PCA)

PCA is a dimensionality-reduction technique. It transforms correlated features into new, mutually orthogonal directions called principal components. The first component captures the greatest possible variance; later components capture as much remaining variance as possible.

• Standardise features when their units or scales differ substantially.

• Compute directions of high variance and project observations onto selected components.

• Use fewer components to support visualisation, compression, or simpler downstream models.

• PCA may reduce interpretability and does not guarantee that the most useful predictive information is retained.

Artificial neural networks

A neural network contains interconnected units arranged in layers. A neuron computes a weighted sum plus a bias and applies an activation function. A multilayer perceptron has an input layer, one or more hidden layers, and an output layer.

Training usually involves a forward pass to compute predictions, a loss function to measure error, backpropagation to calculate gradients, and an optimiser such as gradient descent to update weights. More layers and parameters can model complex patterns but require careful training and sufficient data.

Bias–variance trade-off

A model with high bias is too simple to capture the pattern and may underfit. A model with high variance is overly sensitive to the training sample and may overfit. The aim is good generalisation to unseen data, assessed using a proper validation strategy.

Quick distinction: K-means groups examples; PCA reduces the number of dimensions; a neural network learns layered representations for prediction tasks.

7. Revision Sheet, Applications and Practice

Where ML is used

Area	Possible ML task
Healthcare	Support image analysis or risk prediction, with expert oversight.
Finance	Flag unusual transactions for further review.
Construction	Estimate project costs, forecast delays, or detect safety issues from data.
Agriculture	Classify plant disease from images or estimate crop yield.
Transport	Forecast demand or estimate travel time.

Responsible use of ML

• Protect personal and confidential data; collect only what is needed.

• Check for biased or unrepresentative training data and evaluate performance across relevant groups.

• Explain limitations, uncertainty, and appropriate human oversight.

• Monitor performance after deployment because real-world data can change over time.

Practice questions

1. Define Machine Learning and explain its basic workflow.
2. Differentiate supervised, unsupervised, and reinforcement learning with examples.
3. Write the equation for simple linear regression and explain least squares.
4. What is gradient descent? What is the role of the learning rate?
5. Explain the sigmoid function and logistic regression.
6. Define accuracy, precision, recall, and F1-score.
7. Describe K-means and state two limitations.
8. What is PCA? Explain one benefit and one limitation.
9. Explain overfitting, underfitting, and regularisation.
10. Why must test data remain separate during model selection?

Suggested reading

• KTU B.Tech Curriculum (2024), CSE (AI & ML): introductory AI/Data Science and Concepts in Machine Learning course areas.
• Tom M. Mitchell, *Machine Learning*, McGraw-Hill.
• Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, O'Reilly.

Syllabus note: This is a compact revision resource aligned to introductory KTU B.Tech AI/Data Science and ML themes; it is not an official university publication. Course codes, semester placement, and exact coverage can vary by scheme and programme. Confirm details with the latest KTU syllabus issued by your college.